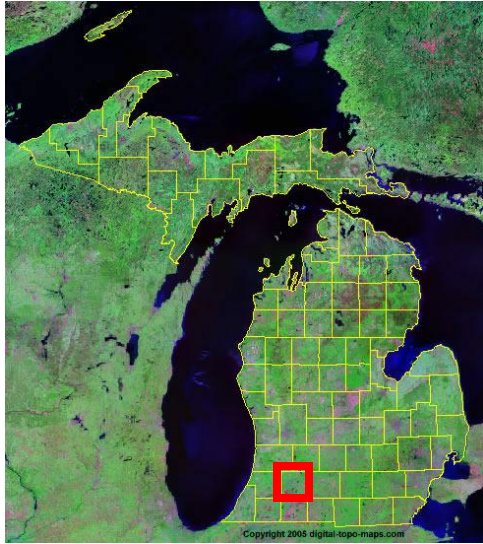


EC 551 – Advanced Digital Design with Verilog and FPGAs

Lecture 1 Introduction to Digital Design

Prof. Douglas Densmore

Who's Teaching This Course??

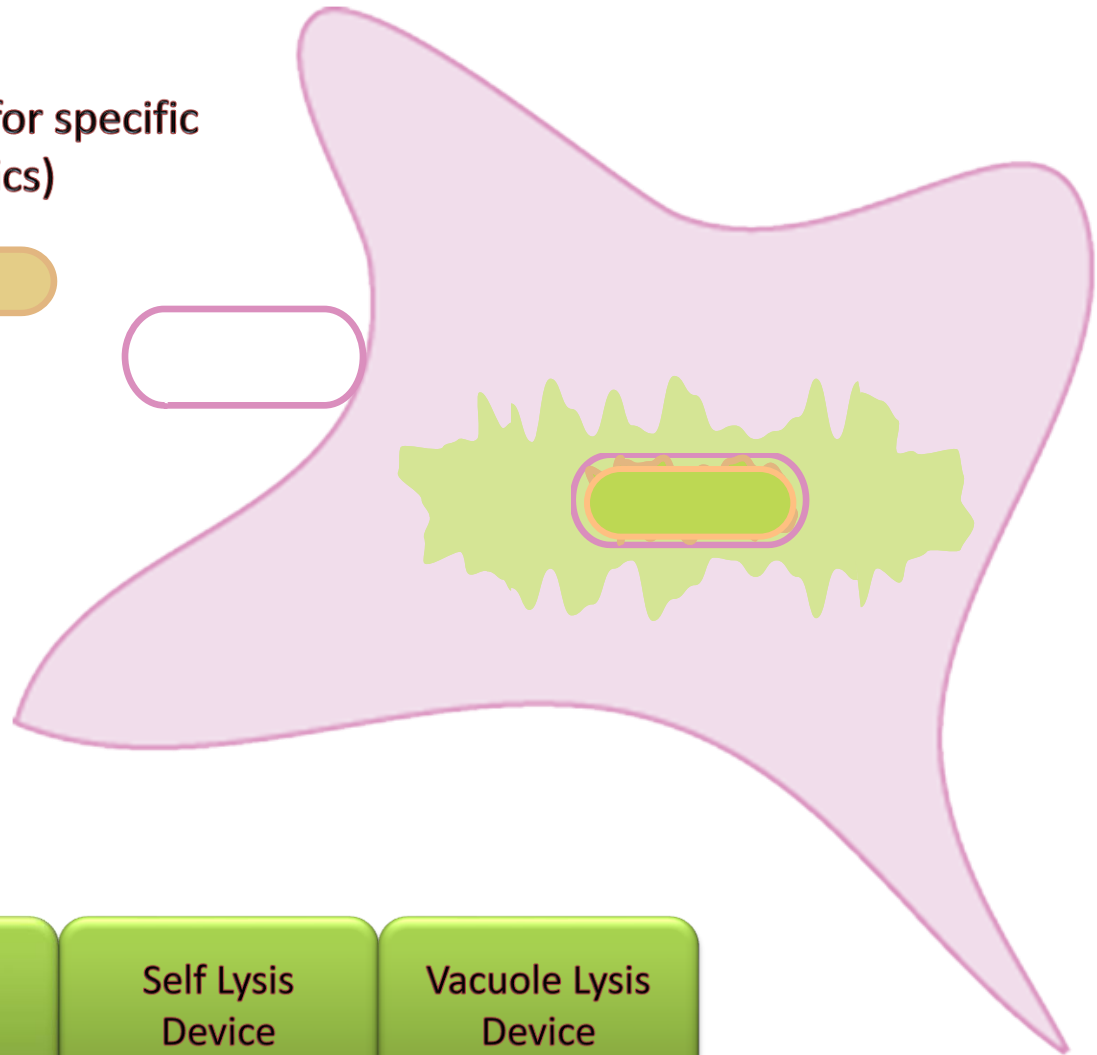




What Do I Do? - General

Synthetic Biology:
Engineering living organisms for specific tasks (example bio-therapeutics)

How do we engineer bacteria to do this using “standard biological parts”? EECS can help!



Invasion
Device

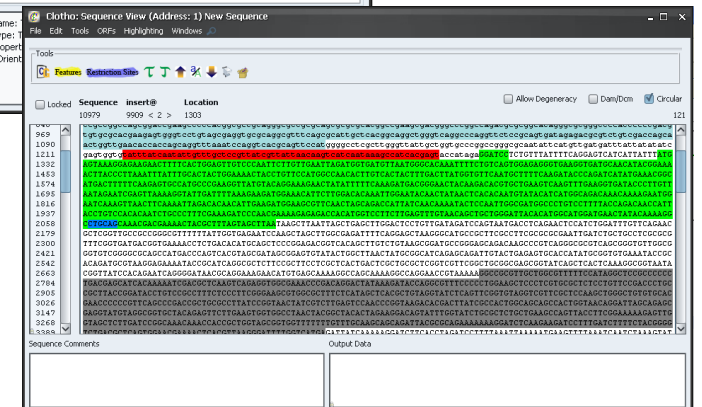
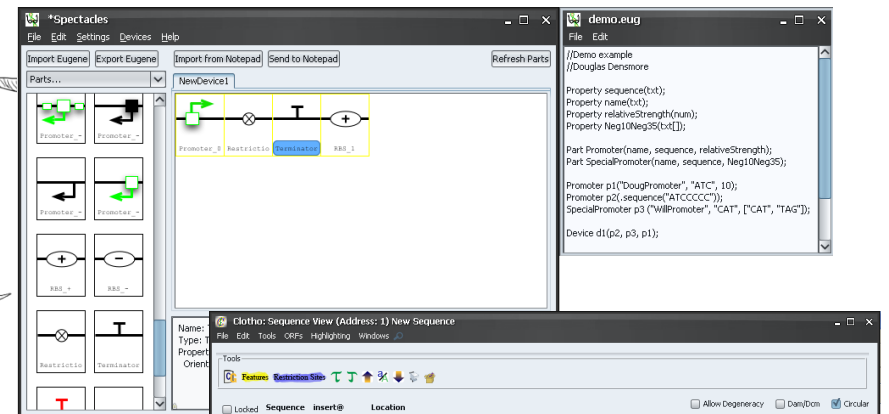
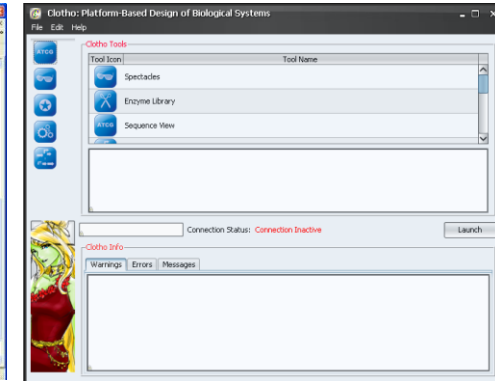
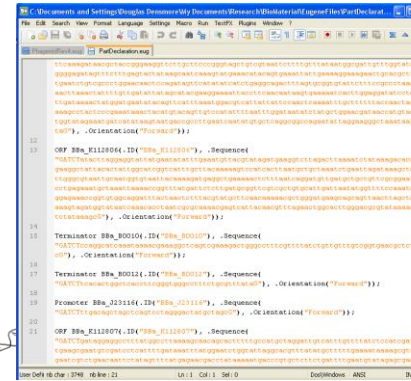
Vacuole
Sensing
Device

Self Lysis
Device

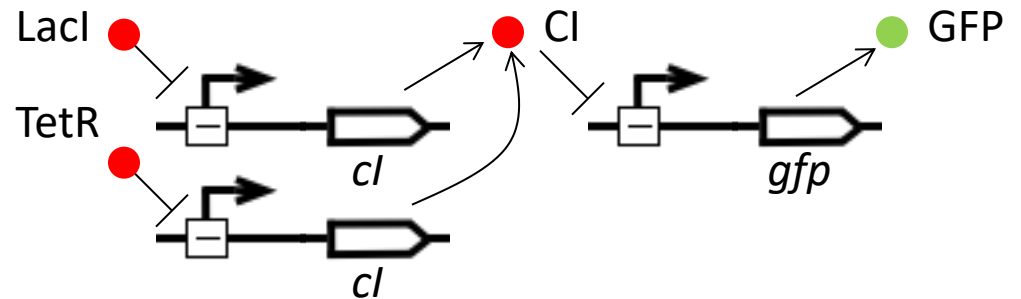
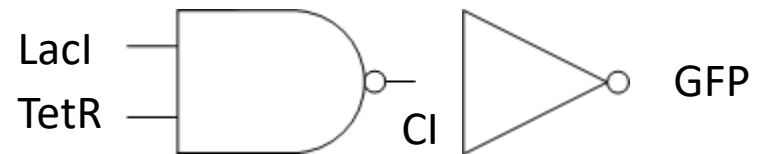
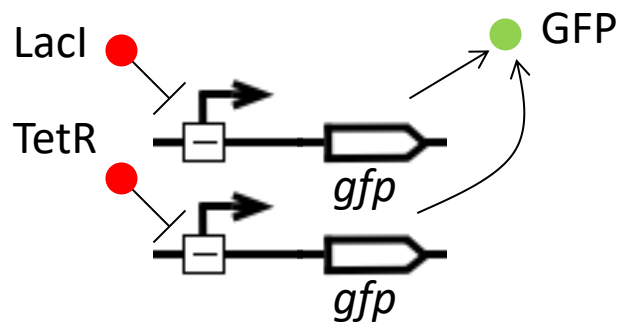
Vacuole Lysis
Device

The logo for Clotho, featuring a stylized white icon of a person with arms raised inside a black rounded square, followed by the word "CLOTHO" in bold, black, uppercase letters.

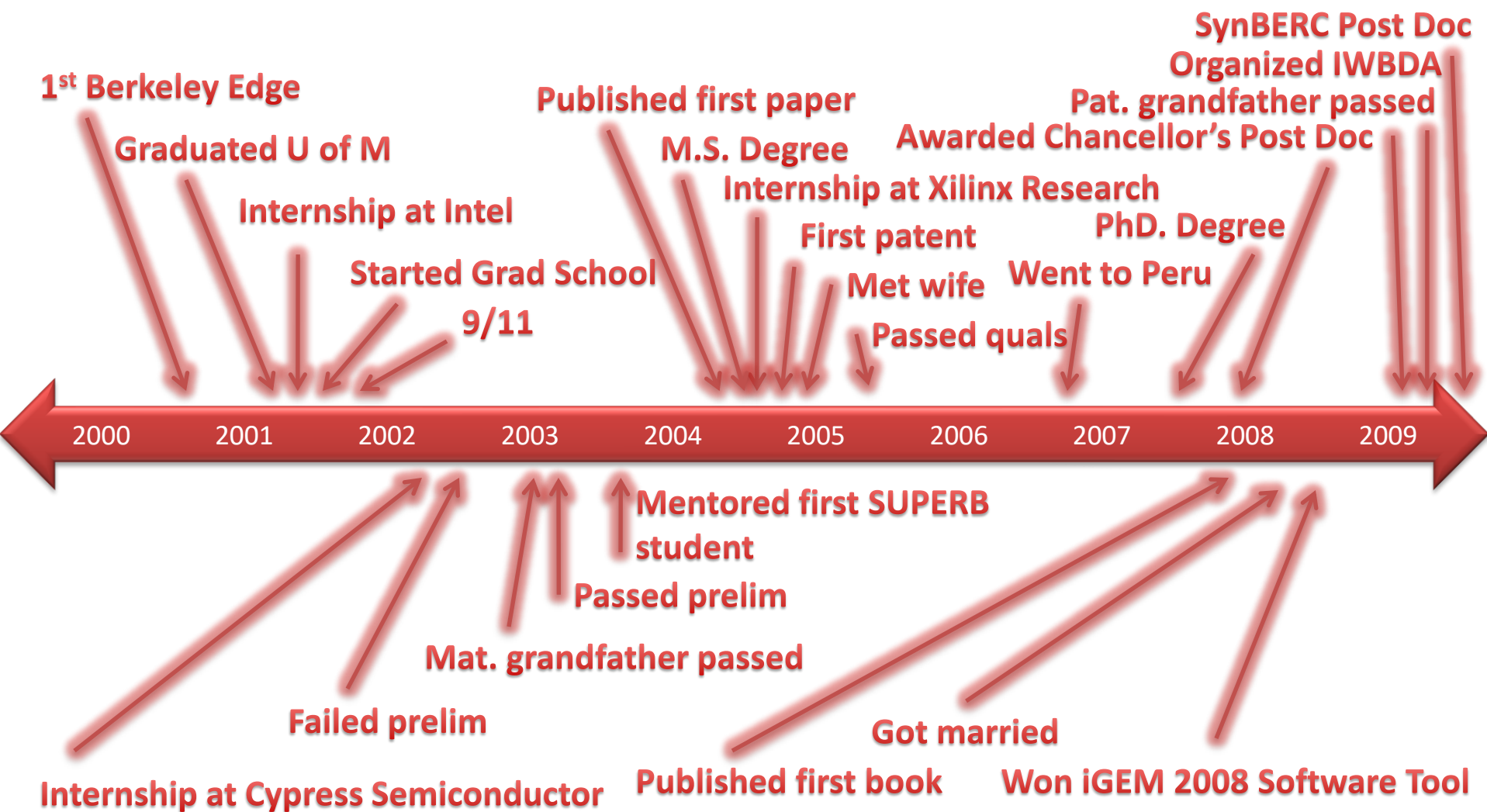
Clotho: Parts Manager (Pro1)												
File Edit Data Tools Help												
Part Nickname	Vector Nickname	Short Description	Format	Author	Composite	Plate	Well	Volume	Status	Label		
- jtk2028	p8ca9145	Ampr, ColE1 Bbb Vector	bbb	Bing_Xia								
- jtk2028	p8ca9145	FLP Recombinase	bbb_1	Josh								
- jtk2028	p8ca9145	FLP Recombinase	bbb_1	Josh		JTK1Plate1	A2	38.0	Sequenced	bcal1109...		
- bcal144	p8ca9145	RFP Cassette	bbb	Josh								
- bcal144	p8ca9145	RFP Cassette	bbb	Josh								
bcal144	p8ca9145	RFP Cassette	bbb	Josh		DougPlat...	D1	0.0	Sequenced	Comple...		
bcal144	p8ca9145	RFP Cassette	bbb	Josh		DougPlat...	B1	0.0	Sequenced	Comple...		
bcal144	p8ca9145	RFP Cassette	bbb	Josh		ChrisPlat	C7	16.0	Functional	bcal144...		
bcal144	p8ca9145	RFP Cassette	bbb	Josh		DougPla...	A3	90.0	Planned	NewLab...		
bcal144	p8ca9145	RFP Cassette	bbb	Josh		DougPlat...	A3	1000.0	Sequenced	NewTest...		
bcal144	p8ca9145	RFP Cassette	bbb	Josh		DougPlat...	B2	0.0	Sequenced	Comple...		
bcal144	p8ca9145	RFP Cassette	bbb	Josh		DougPlat...	D3	1000.0	Sequenced	NewTest...		
bcal144	p8ca9145	RFP Cassette	bbb	Josh		DPlate2	A1	0.0	Sequenced	Hello		
bcal144	p8ca9145	RFP Cassette	bbb	Josh		DougPlat...	A4	1.0	Sequenced	NewSam...		
bcal144	p8ca9145	RFP Cassette	bbb	Josh		TestPlat...	B3	10.0	Functional	NewSam...		



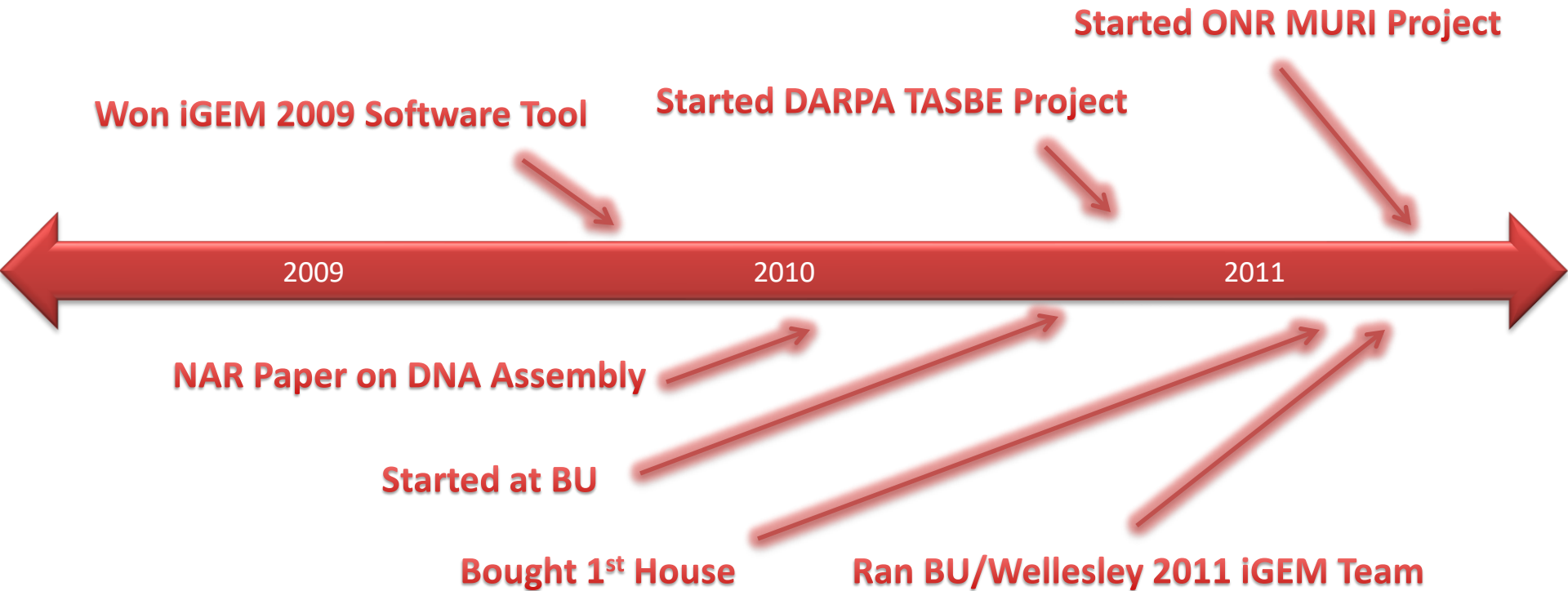
Why Am I in the ECE Dept?



How I Got Here (Short Version)



Timeline Extended



What is this course about?

- Let's review the syllabus together
 - Available at <http://learn.bu.edu>
 - Turn in “Syllabus Confirmation” form by 9/15
- Understand what the programs/project **is** and what it **isn't**
 - Isn't all figured out in advance.
 - Is a “place” where you will get the answer.
 - Isn't “extra”.
 - Is the main way you are going to really learn the material.
 - Is about getting things to work. Isn't about getting things to work. Huh??

General Overview of the Course

- Prof. Densmore is not a Verilog expert BUT together we WILL find the solution.
- Practice, practice, practice, and then...practice.
- Collection of **programs** with HW to serve as check points for the exams.
 - Exams serve as a way for Prof. Densmore to differentiate students and identify students that are struggling.
- **Project!!!!!!**

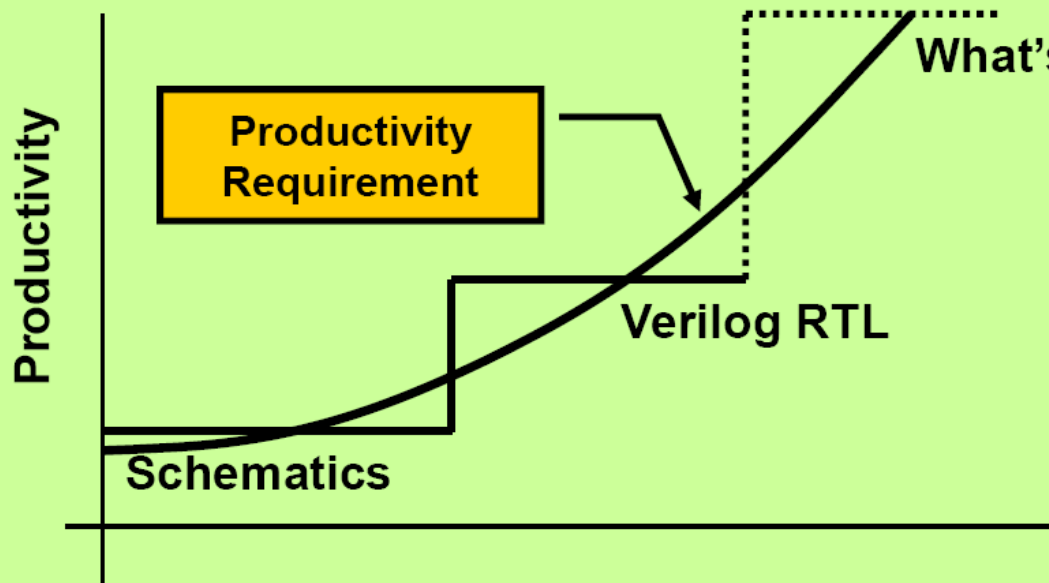
Why Digital Design?

<https://www.youtube.com/watch?v=MEDmu61Bdzg>

Discussion?

Videos!!!

Verilog RTL Today is Where Schematics Were 15 Years Ago

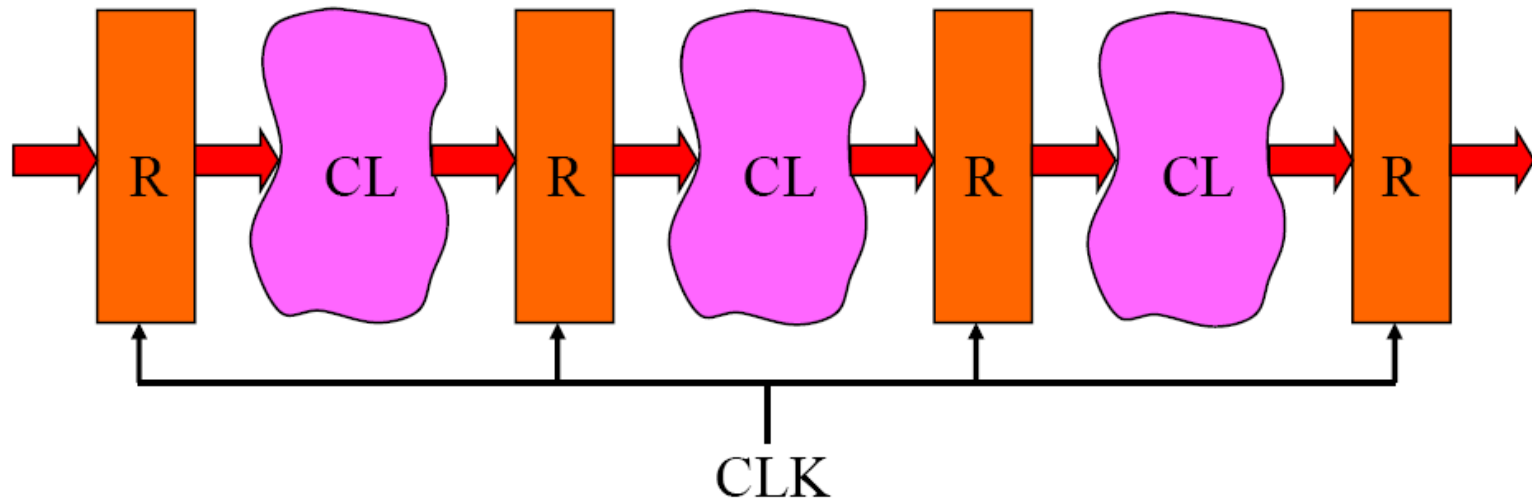


What's Next?- See at the end of EC551



RTL Design

- Big simplification of synchronous circuit design process
- Separation of optimization and timing
 - All sequential part – interaction between registers
 - All synthesis is about combinational clouds



Design Flow Overview

Design and implement a simple unit permitting to speed up encryption with RC5-similar cipher with fixed key set on 8031 microcontroller. Unlike in the experiment 5, this time your unit has to be able to perform an encryption algorithm by itself, executing 32 rounds.....

Specification

Verilog or VHDL description

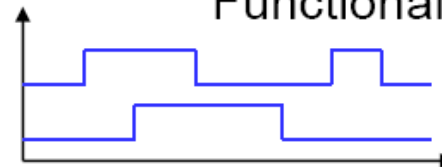
```
Library IEEE;
use ieee.std_logic_1164.all;
use ieee.std_logic_unsigned.all;

entity RC5_core is
  port(
    clock, reset, encr_decr: in
    std_logic;
    data_input: in
    std_logic_vector(31 downto 0);
    data_output: out
    std_logic_vector(31 downto 0);
    out_full: in std_logic;
    key_input: in
    std_logic_vector(31 downto 0);
    key_read: out std_logic;
  ); <...>
End RC5_core;
```

Simulation



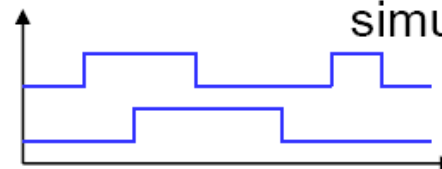
Functional simulation



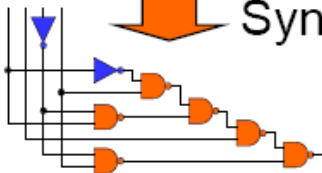
Simulation



Post-synthesis simulation



Synthesis



Design Flow Overview

- Overview of the design flow
 1. Design specification
 2. Design partition
 3. Design entry
 4. Simulation/functional verification
 5. Design integration and verification
 6. Presynthesis sign off
 7. Synthesize and map gate-level netlist
 8. Postsynthesis design validation
 9. Postsynthesis timing verification
 10. Test generation and fault simulation
 11. Cell placement, scan chain and clock tree insertion, cell routing
 12. DRC
 13. Extract parasitics
 14. Design sign-off

Not covered in this class in depth

Design Specification

- Functional characteristics
 - State transition graphs
 - Timing charts
 - Algorithmic-state machines
- SystemC, System Verilog, Superlog ???
- NOT THE VERILOG CODE

Discuss

Design Partition

- Create a configuration of interacting functional units.
- This is the **architecture**
 - What is the microarchitecture?
- This process is often called “top down” or “hierarchical design”

Discuss

Design Entry

- Language based description of the design
 - THIS IS COULD BE VERILOG
- Behavioral vs. Structural modeling
 - Understand this!!!
- Mixed abstraction level support ideally

Discuss

Simulation and Functional Verification

- Simulation vs. formal methods
 - Test plan development
 - Testbench development
 - UUT?
 - Test execution and model verification
- Formal vs. simulation explain??!!
- When are you done?

Discuss

Design Integration and Verification

- Piece together the design
 - Now verify the whole system
 - Another testbench!!

Discuss

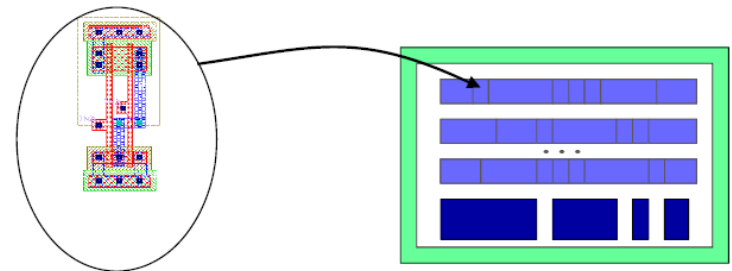
Presynthesis Sign-Off

- Behavior model must match the design spec.
- Who signs this off?

Discuss

Gate-Level Synthesis and Technology Mapping

- Assign physical resources (standard cells, LUTs, etc) to boolean algebraic expression (as a netlist).
- This is a HUGE step
 - Don't mistake this just for one slide 😊



Discuss

Postsynthesis Design Validation

- Compare behavior description to synthesized gate-level description
- How do you compare?
 - Common stimulus “Y-chart” approach

Discuss

Postsynthesis Timing Verification

- Take into account capacitive delays and geometric details of the layout.
- May require
 - Transistor sizing
 - Architectural modifications/substitutions
 - Device substitution

Discuss

Test Generation and Fault Simulation

- Application of “test vectors”
 - ATPG??
- Combinational logic vs. sequential logic
 - Challenges?
- Tests are produced before device is fabricated.

Discuss

Placement and Routing

- Place cells on the die and connect their signal paths.
 - Why are these done together?
 - Local vs. global routing?
 - Channel vs. switchbox routing
- Clock tree generation

Discuss

DRC

- Physical layout rules
 - Widths, overlaps, separations
 - What are some common features?
 - Metal, vias, connects, poly, wells, etc.
- Electrical rules
 - Fan-in, fanout, signal integrity

Discuss

Parasitic Extraction

- Parasitic capacitance used for more accurate verification.
- Update timing models.
 - Here we go again 😊
 - Classic place to experience **“timing closure”** issues

Discuss

Design Sign-Off

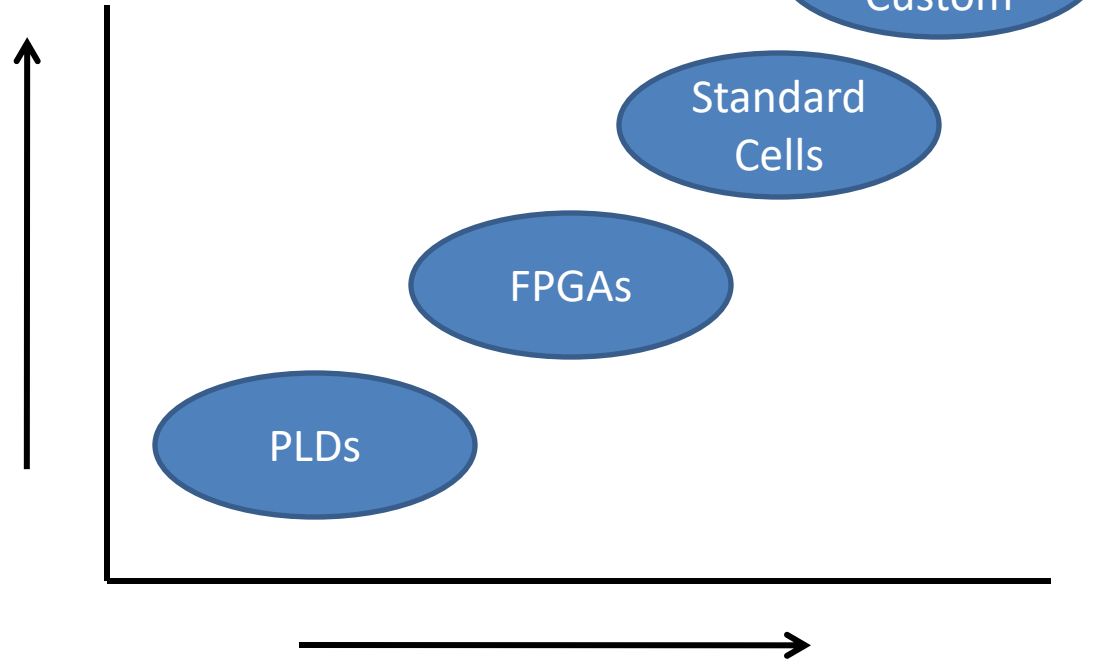
- Design constraints must be met
- Final timing closure
- Produce final mask set (GDS-II)
 - Binary file format representing planar geometric shapes, text labels, and other information about the layout in hierarchical form
- **Done!!!!**

Discuss

IC Technology Options

- PLDs
- FPGAs
- Standard Cells
- Full Custom IC

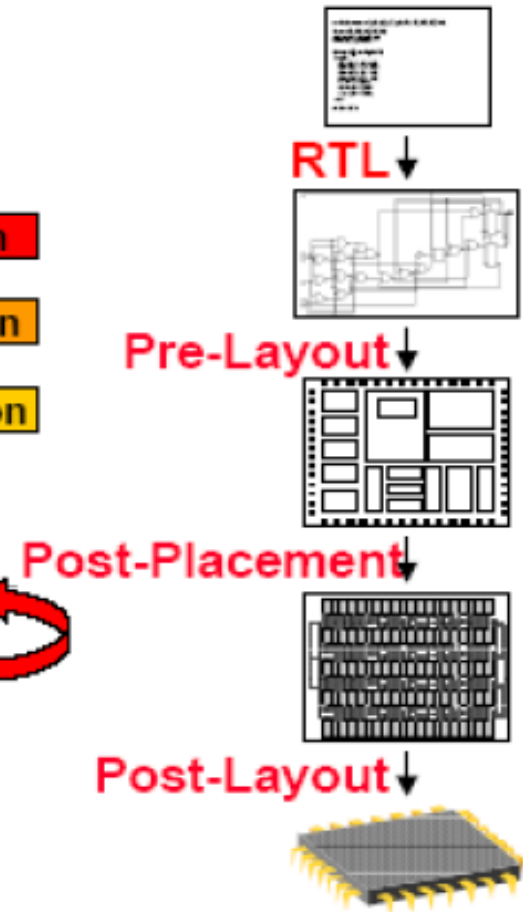
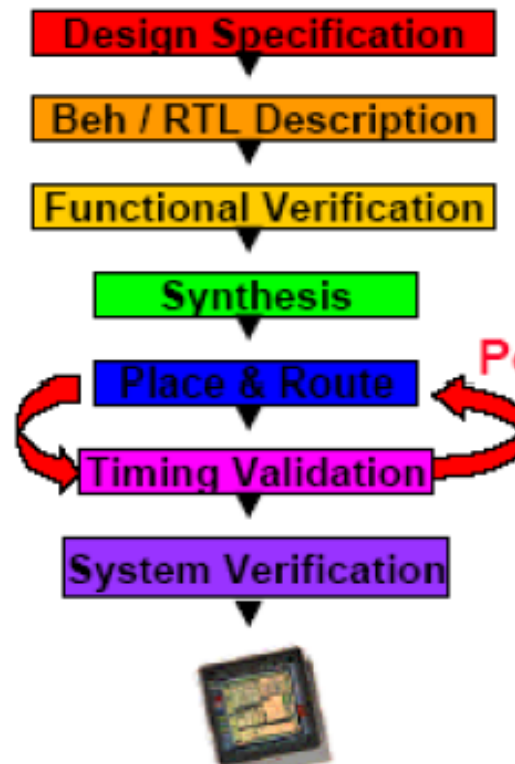
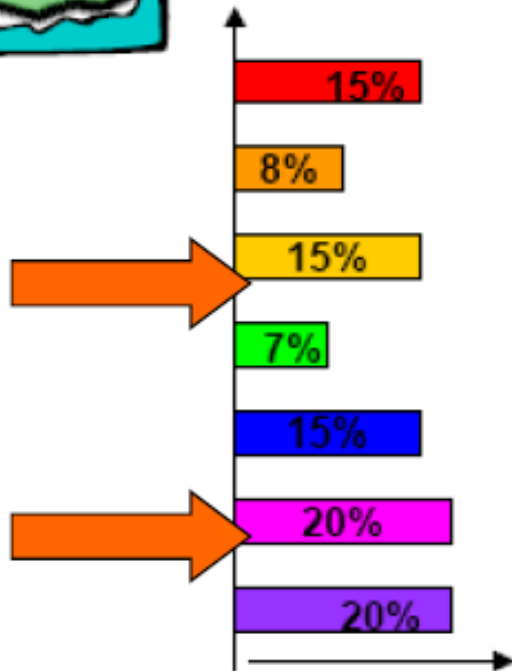
Time to prototype
Market to volume amortize



Discuss

NRE costs
Process complexity
Density, Speed, Complexity

A Designer's Life



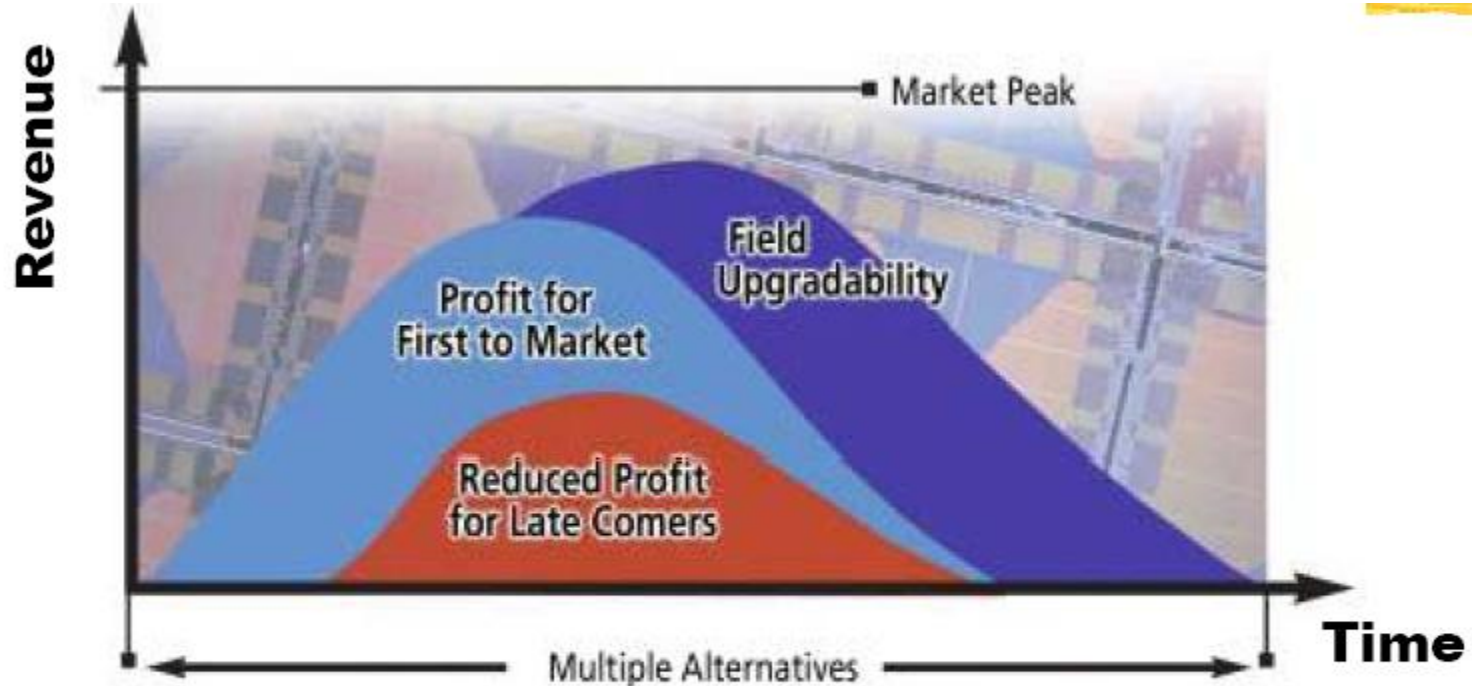
Market Conditions

- ❑ 18% of all projects are cancelled within 5 months*
- ❑ 58% are late to market*
- ❑ 20% of products are not within 50% of specification*
- ❑ Of the products that do get to market:
 - On time and 50% over budget earn only 4% less profit over 5 years†
 - 6 months late and on budget earn 33% less profit over 5 years†

* Source: *Current and Emerging Embedded Markets and Opportunities*

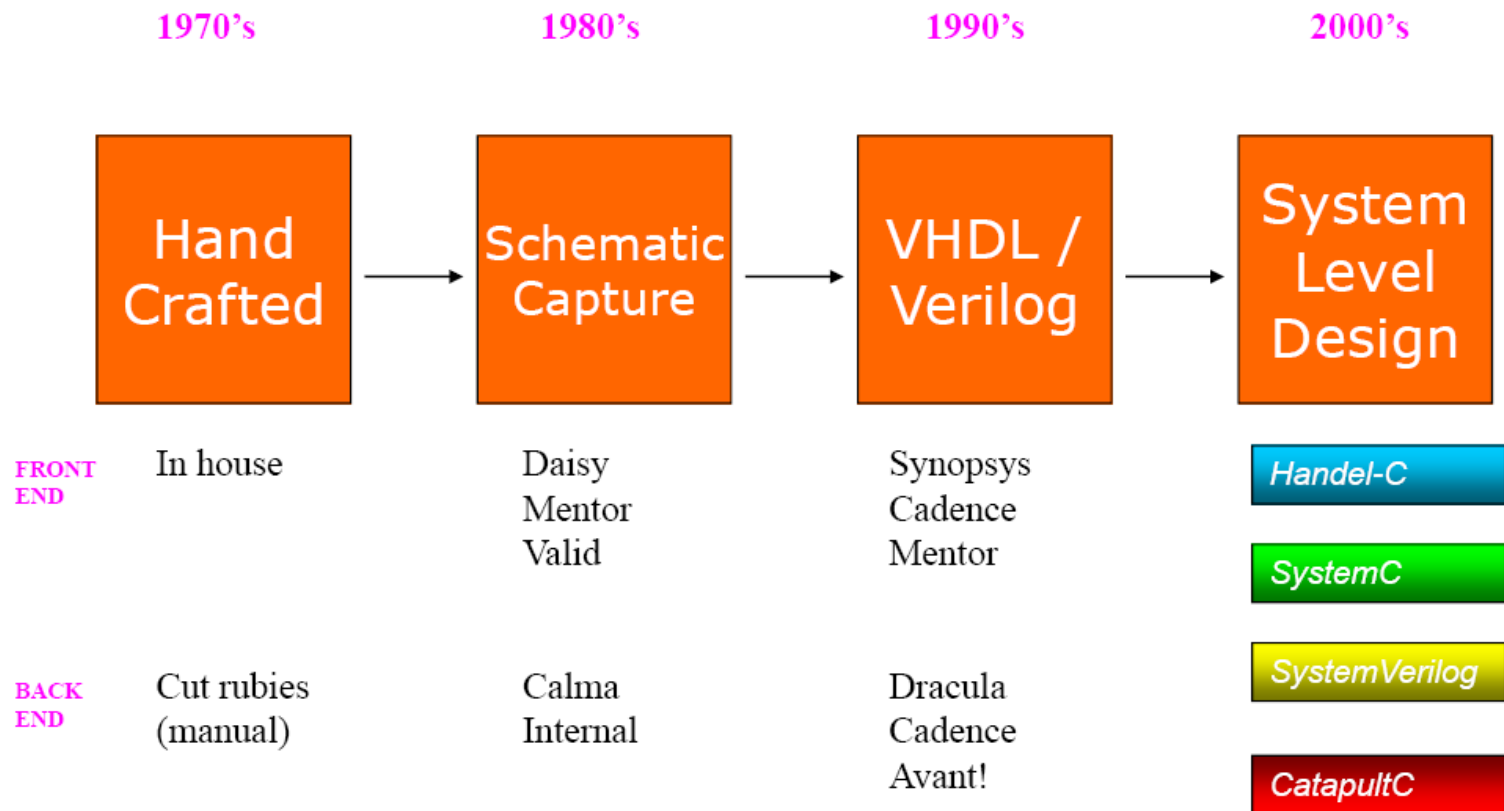
† Source: *McKinsey & Co.*

Impact of the Marketplace

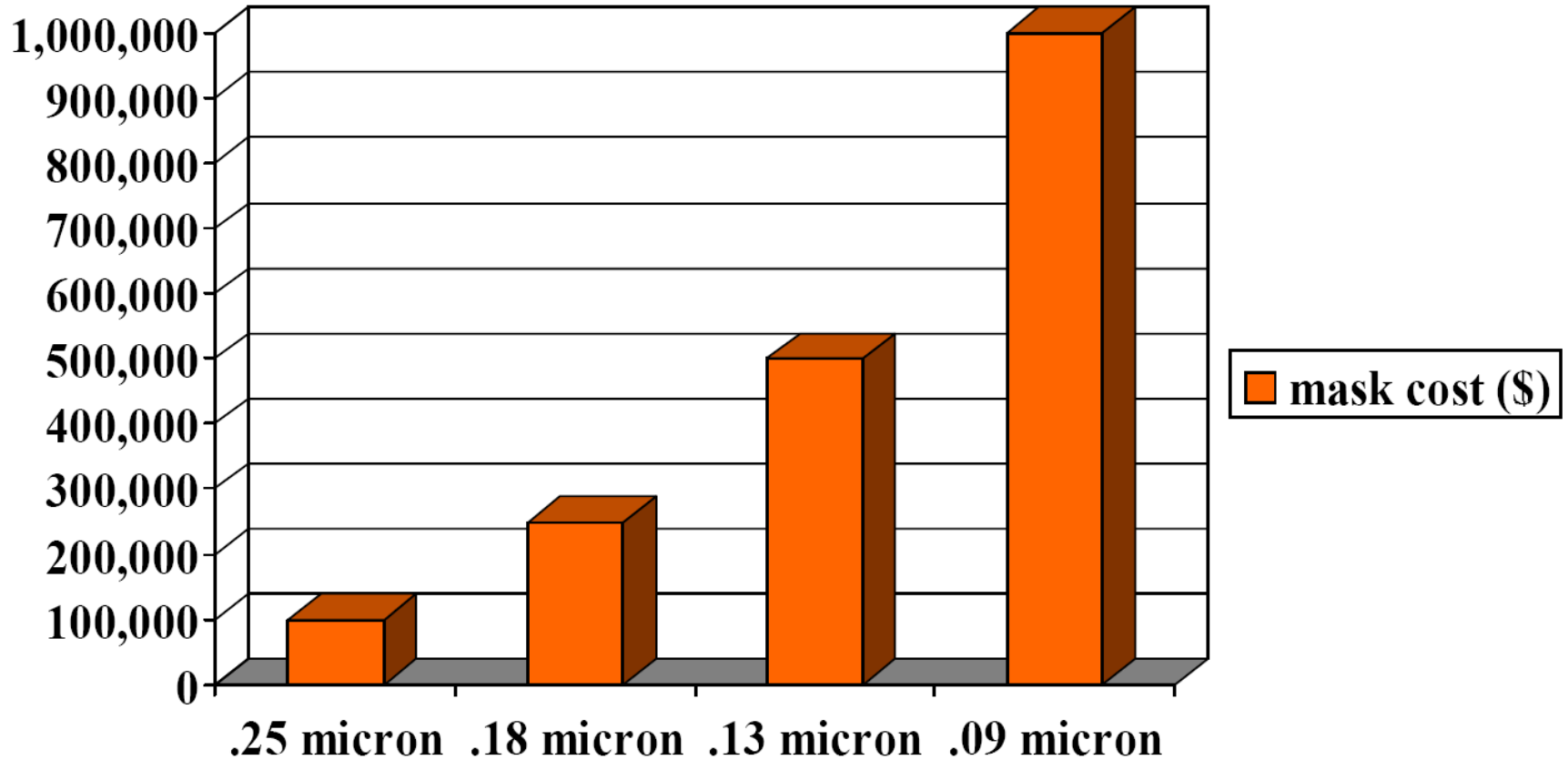


- First 10%-20% of the design cycle can set 70%-80% of system cost
- 4-week product rollout delay results in a 14% loss in market share
- Pressure is to shorten the design cycle

Semiconductor Design

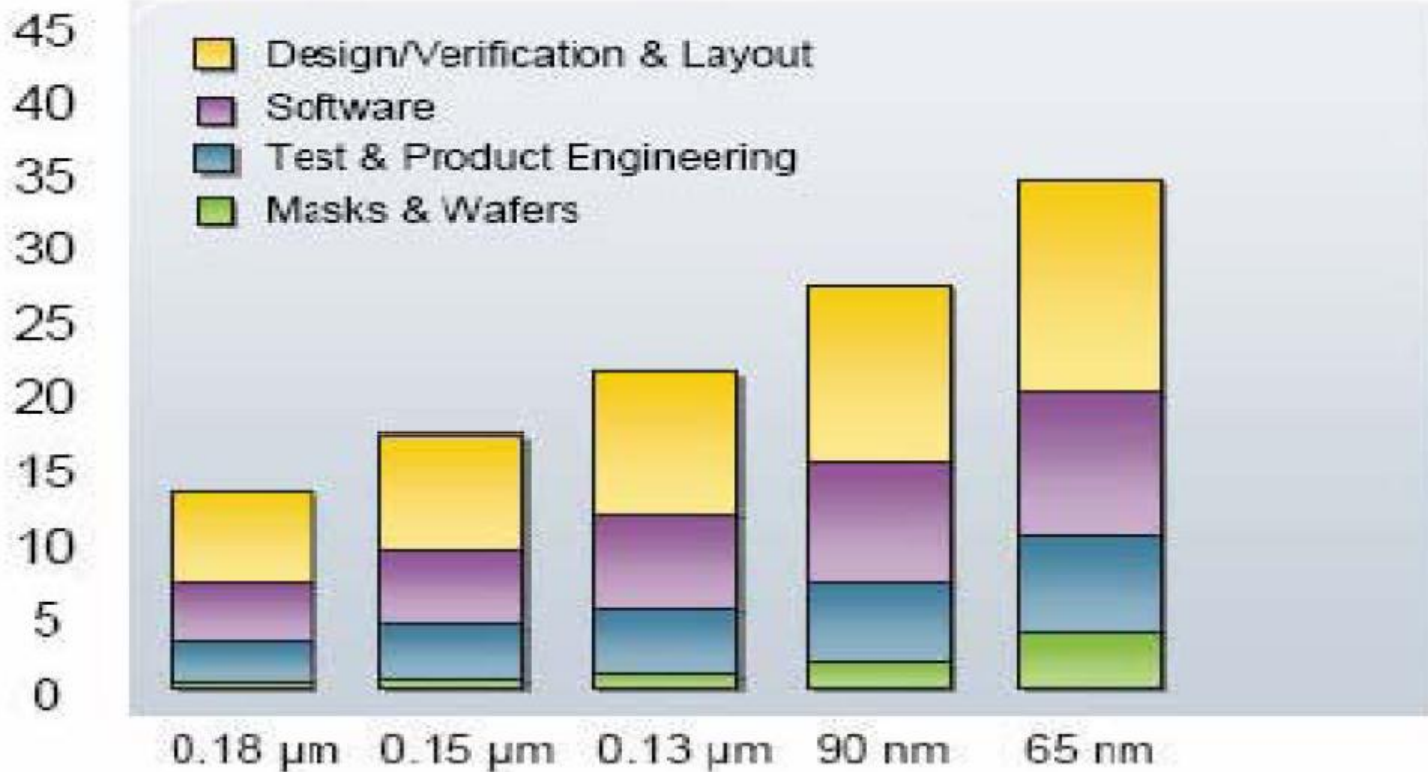


Mask Cost Vs. Line Width



Development Costs

Total Development Costs (\$M)



Source: IBS (March 2005)

ASICs Costs!!

Mask-set cost: ~ \$1M

⇒ NRE (**non-recurring engineering charge**) > \$3M

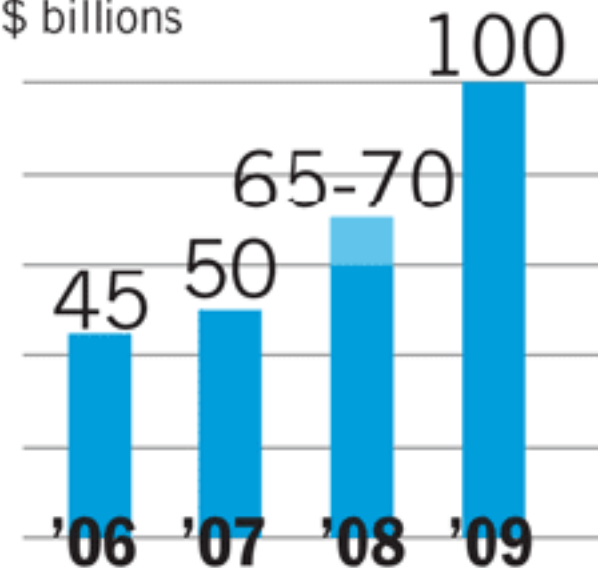
⇒ Justifying production value > \$10M



**Less than 1000 ASIC designs out of 10,000
could justify full mask-set cost at 0.13μ**

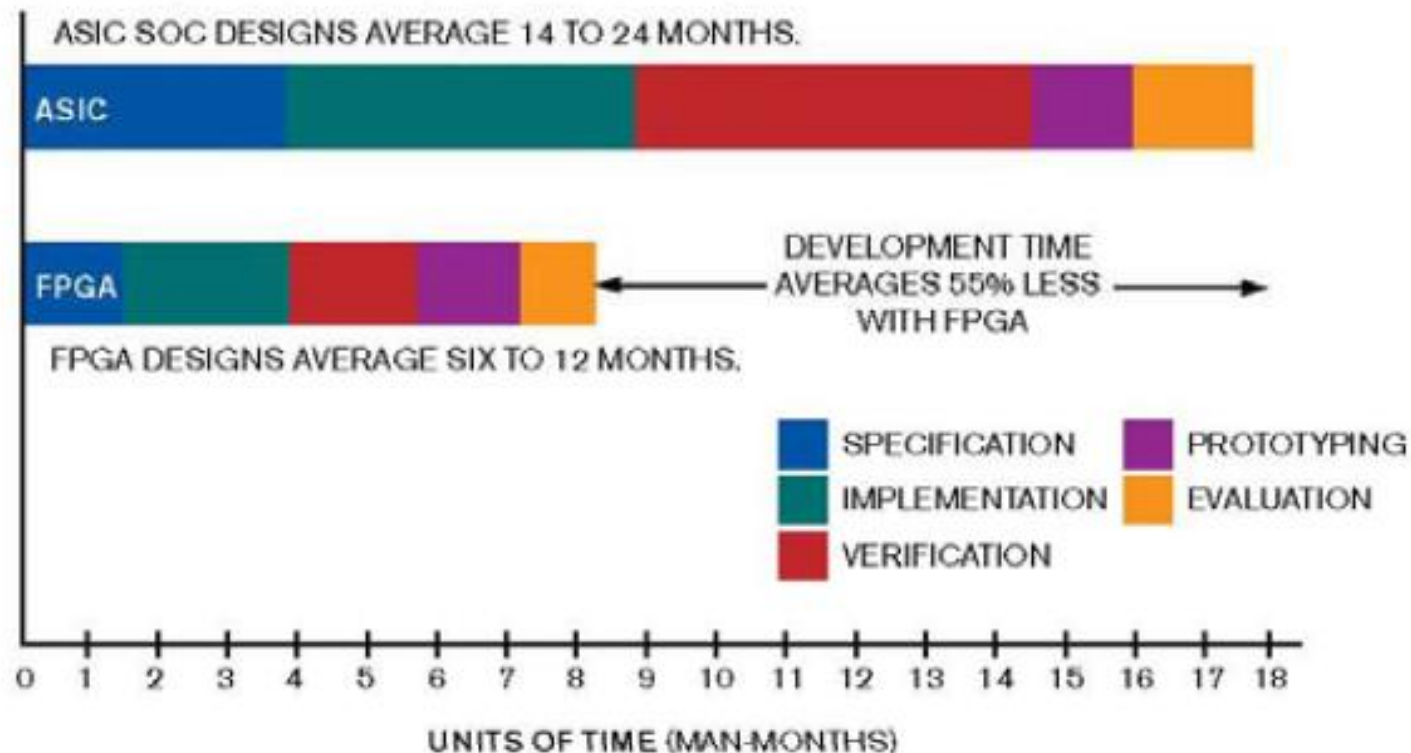
Total cost of semiconductor R&D

Can anyone afford it?
\$ billions



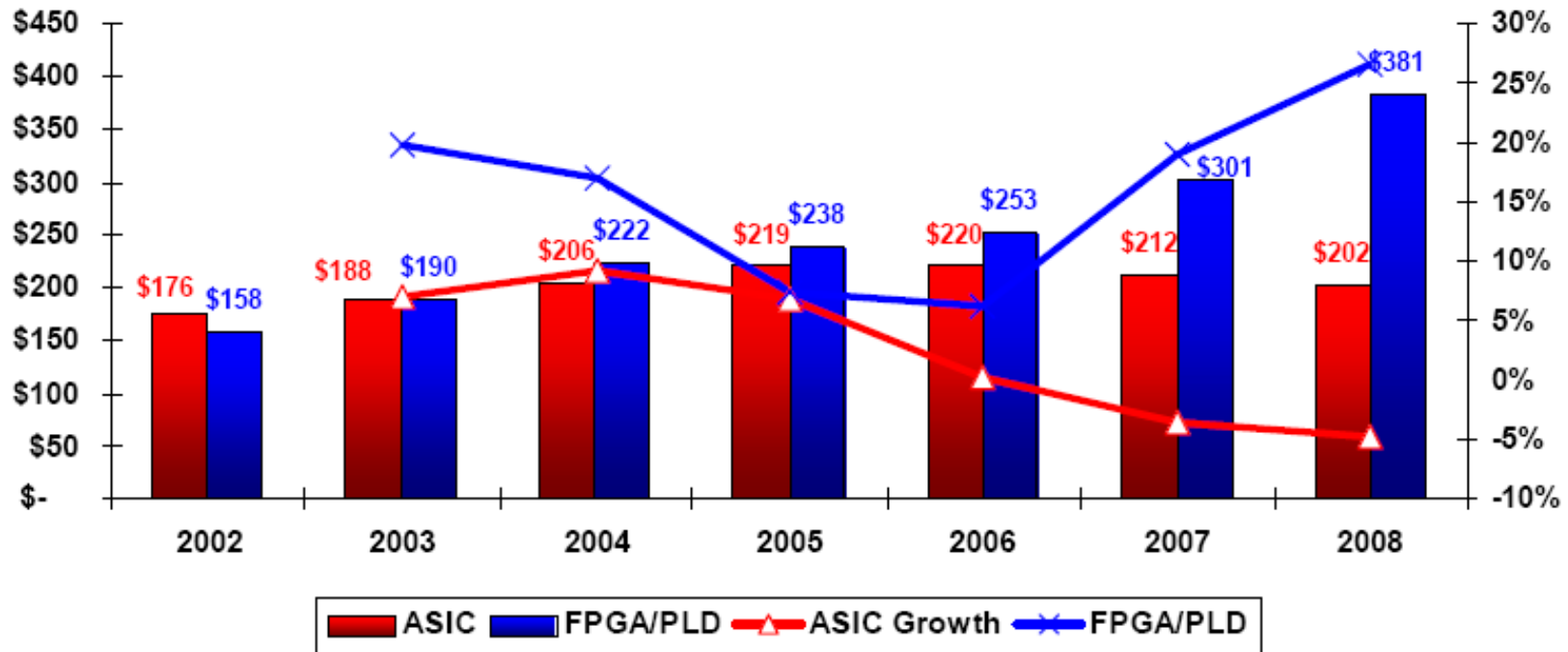
SOURCE: Intermolecular

ASIC vs. FPGA Design Cycle



SOURCE: *EXPERIMENTATION MATTERS: UNLOCKING THE POTENTIAL OF NEW TECHNOLOGIES FOR INNOVATION*, STEFAN H THOMKE, HARVARD BUSINESS SCHOOL PRESS, 2003. BASED ON A HARVARD STUDY OF 400 DESIGNS.

FPGAs in Aero/Mil Segment



Source: Gartner Dataquest

FPGA Development



Xilinx FPGAs



- Virtex-6 and Spartan-6 FPGAs build on the success of the Virtex-5 family.
- The **Virtex-6** family is built on a **40-nm** process for compute-intensive electronic systems, and the company claims it consumes 15 percent less power and has 15 percent improved performance **over competing 40 nm FPGAs**
- **The Virtex-6: 760,000 logic cells, 948,000 flip-flops**, and 38 Mb of block RAM, compared to 360,000 logic cells, 207,000 flip-flops, and 18 Mb of block RAM in the largest Virtex-5 FPGA
- The **Spartan-6** family is built on a **45-nm**, 9-metal layer, dual-oxide process technology.
- The Spartan-6 is a low-cost solution for automotive, wireless communications, flat-panel display and video surveillance applications
- The eleven **Spartan-6** devices cover a wide range, from 3,800 to **147,000 Logic Cells**, with **up to 184,000 flip-flops** and up to 4.8 Mb of internal block RAM

ASICs vs. FPGA Recap

- We are using FPGAs in this class

WHY??

Next Time

- Download the combinational logic and sequential logic “quizzes”
- Overview of combinational logic (in one lecture!)
- Read Chapter 1 – Not long. It is a review of these slides.
- Read chapter 2 – we are going to go through it all.

Questions?